



CS766 Computer Vision
Final Project

Pedestrian Intent Prediction for Autonomous Vehicles

VLM-Guided Scene Context for Intent Prediction

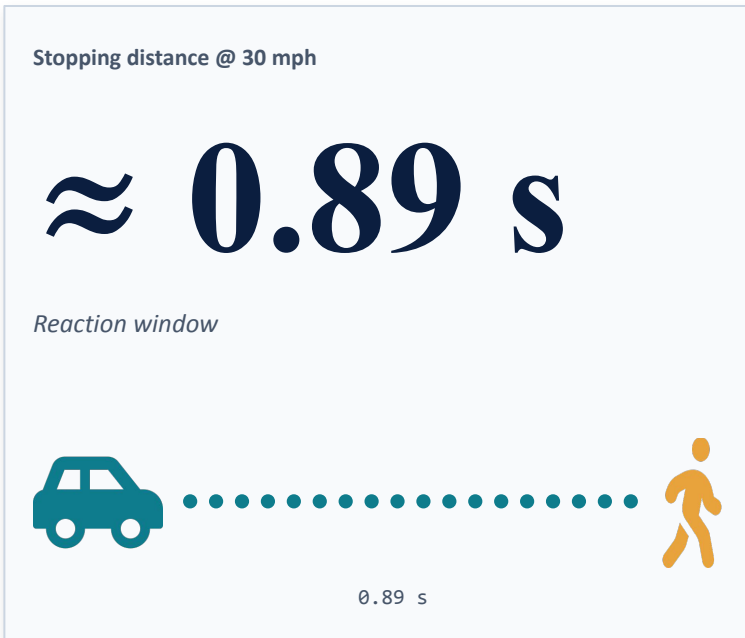
Aditya Kumar · Amogh Dixit · Jahnavi Sunchu



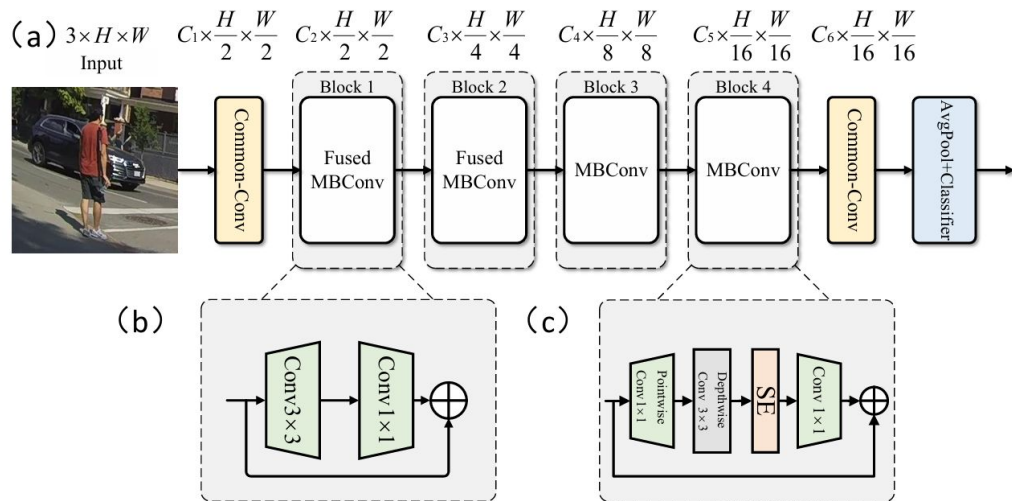
Anticipating, not reacting.

A pedestrian who steps off the curb is already inside the vehicle's stopping distance.

Pedestrian intent prediction: given a single observation of a pedestrian, predict, in real time, *will cross* or *won't cross*.



EfficientPIE (IJCAI 2025)



0.21 ms

inference per frame



89%

reported accuracy on JAAD



no temporal

no recurrence, no scene context

Accuracy can mislead.



EfficientPIE (IJCAI 2025) reports 89% accuracy on JAAD at 0.21 ms/frame inference.

But the test set is heavily unbalanced, with way more non-crossing examples (1584) than crossing examples (292).



Hidden failure modes

A model perfect on easy cases and consistently failing on hard ones can still report high overall accuracy.



Confidence $\neq$ correctness

If wrong predictions come with high confidence, downstream safety logic has no signal to fall back on.

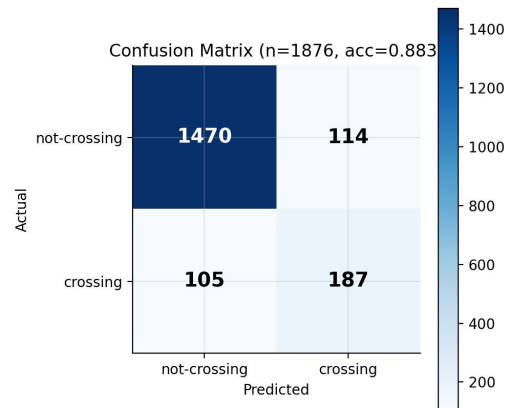


Our question

When EfficientPIE is wrong, where does it go wrong and does it know that it's wrong?

Baseline Reproduction

*Reproduced the paper's training pipeline on JAAD:
88.3% accuracy but **64% recall**.*



DEMOGRAPHIC

Who does it fail on?

Stratify errors by age, gender, group size.



SCALE

How does distance change things?

Stratify errors by pedestrian bounding-box area as a proxy for how close the pedestrian is.

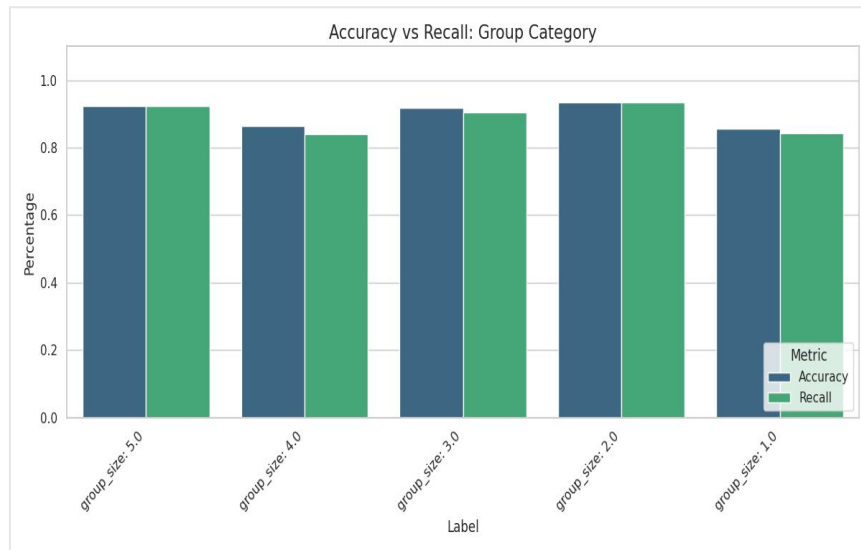
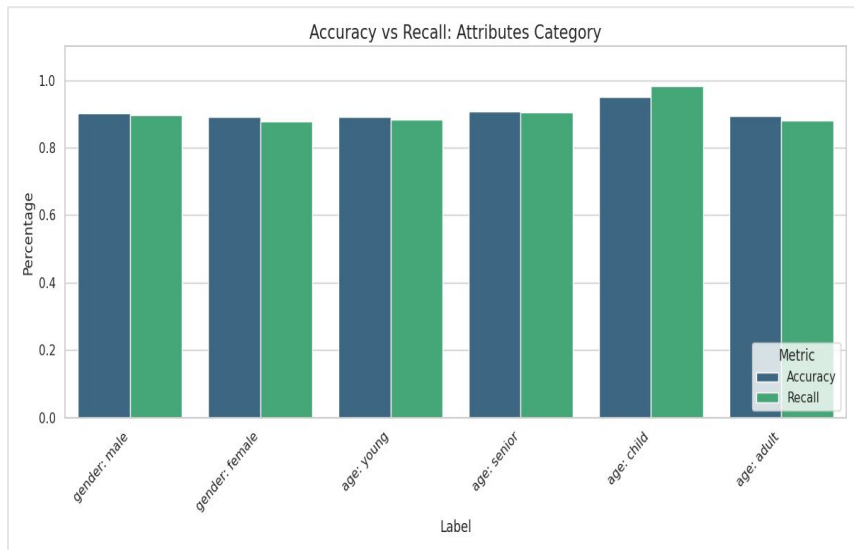


OCCUSION

What happens when context is blocked?

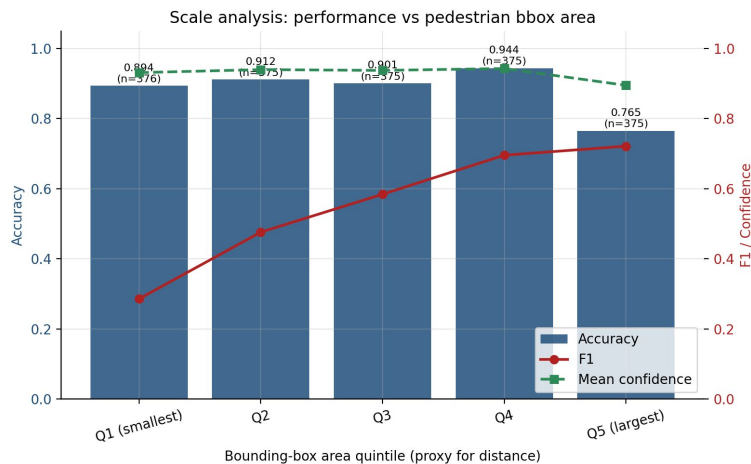
Compare clean, partial, and full-occlusion cases. Use GradCAM to see what the model attends to.

Demographic bias analysis



We first tested for demographic bias, stratifying the model's errors across age, gender, group size, and motion direction. While the charts show slight visual variances in accuracy and recall, none of these gaps are statistically significant. The model simply doesn't fail based on *who* the pedestrian is. Ruling out demographics pointed us directly toward environmental factors instead, specifically, how close the pedestrian is to the camera.

Scale Paradox

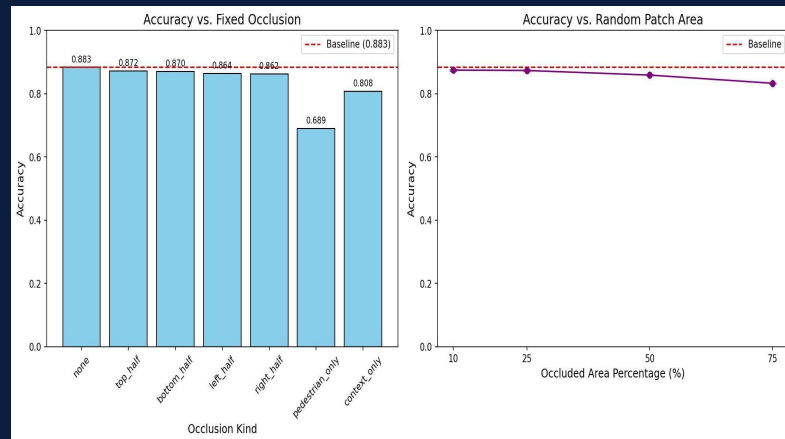


Performance severely degrades when pedestrians are closest to the camera (Q5), with accuracy plummeting to 76.5%

The model makes errors on the closest pedestrians with high confidence. This indicates that the model struggles with lack of surrounding environmental context

Spatial Ablation Analysis

We applied 11 masking conditions to every test image and measured how performance changed per condition.



- Half-body masks — top, bottom, left, right halves
- Random patches — 10%, 25%, 50%, 75% coverage
- Pedestrian only — entire background removed
- Context only — pedestrian removed entirely

Accuracy stays near baseline for most conditions but accuracy hides the real story.

Spatial Ablation Results

0.064

F1 — pedestrian_only

near chance

0.457

F1 — context_only

model still works

0.631

F1 — none (baseline)

full input



Grad-CAM++: Where Is The Model Looking?

Red = model attention. Blue = ignored. The pedestrian body is cold in every image.



Storefront scene

Attention on entrance, parked car, curb boundary. Pedestrian body cold.



Road crossing

Crossing pedestrian ignored. Attention scatters to background houses and nearby road.



Child mid-cross

Model locked onto SUV and taxi sign. Child completely invisible to the model.

Two findings, same cause

SCALE PARADOX

Accuracy goes down as the pedestrian becomes more visible.

→ the missing signal isn't in the crop.

GRADCAM

The model attends to background context, not the pedestrian.

→ scene context is already what's driving predictions.



The model relies on scene context but it's doing so opportunistically.

It learns brittle, video-specific scene patterns rather than principled scene reasoning.

Hypothesis: Use CLIP-VLM as offline teacher on full uncropped frames → force student to learn global scene semantics → zero inference cost change

VLM Knowledge Distillation

Architecture

Student	EfficientPIE (frozen backbone)
Teacher	CLIP ViT-B/32 (offline, full scene)
Projector	1280 → 1024 → 768 (MLP)
Hook	model.flatten [B, 1280]

Offline Pipeline

- 1 CLIP processes full uncropped JAAD frames
- 2 768-dim embeddings extracted per sample
- 3 Embeddings saved to disk alongside crops
- 4 Loaded as 3rd tensor during training

Loss Function

$$L = L_{\text{task}} + \alpha \times L_{\text{distill}}$$

L_{task}	CrossEntropyLoss (crossing / not-crossing)
L_{distill}	MSELoss (student projector vs VLM embedding)
α	0.5 — equal task / distillation weight

VLM Knowledge Distillation

Baseline Model



After Distillation



Result: Accuracy, F1 and Recall unchanged. Heatmaps show more concentrated background attention: sharper locks on rooflines and architecture. Pedestrian still cold.

Future Work

01 Scene-Balanced Dataset

Ensure each scene type appears in both crossing and non-crossing conditions. Decorrelates filming location from the crossing label forcing the model to find genuine behavioral signal rather than exploiting location shortcuts.

02 Pedestrian-Grounded VLM Distillation

Replace CLIP with a domain-aware VLM like LLaVA, prompted on pedestrian-specific questions: "Is this person at a crosswalk?", "Are they facing the road?", "Are they near a curb?". CLIP was trained on general image-text pairs with no crossing-specific exposure, its embeddings could be satisfied by background scene templates alone. A pedestrian-grounded teacher produces a signal the student cannot satisfy without attending to the pedestrian, preserving real-time inference.

THANK YOU